ALICE L WALTON SCHOOL OF MEDICINE

Supportive Therapy for Patients on Chemotherapy – Antiemetics

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of the antiemetic agents.
2. Identify adverse effects associated with the antiemetic agents.

CASE CONNECTION

RW is a 58-year-old female you are seeing in the oncology outpatient clinic. RW has breast cancer, has ongoing chemotherapy for her cancer, and chronic pain secondary to bony metastases. Her pain is well controlled on nonsteroidal anti-inflammatory drugs and low-dose opioids. “I can deal with the pain,” RW says. “I just feel so nauseous. Is there anything I can try?” She currently has a prescription for ondansetron and has been taking it consistently.

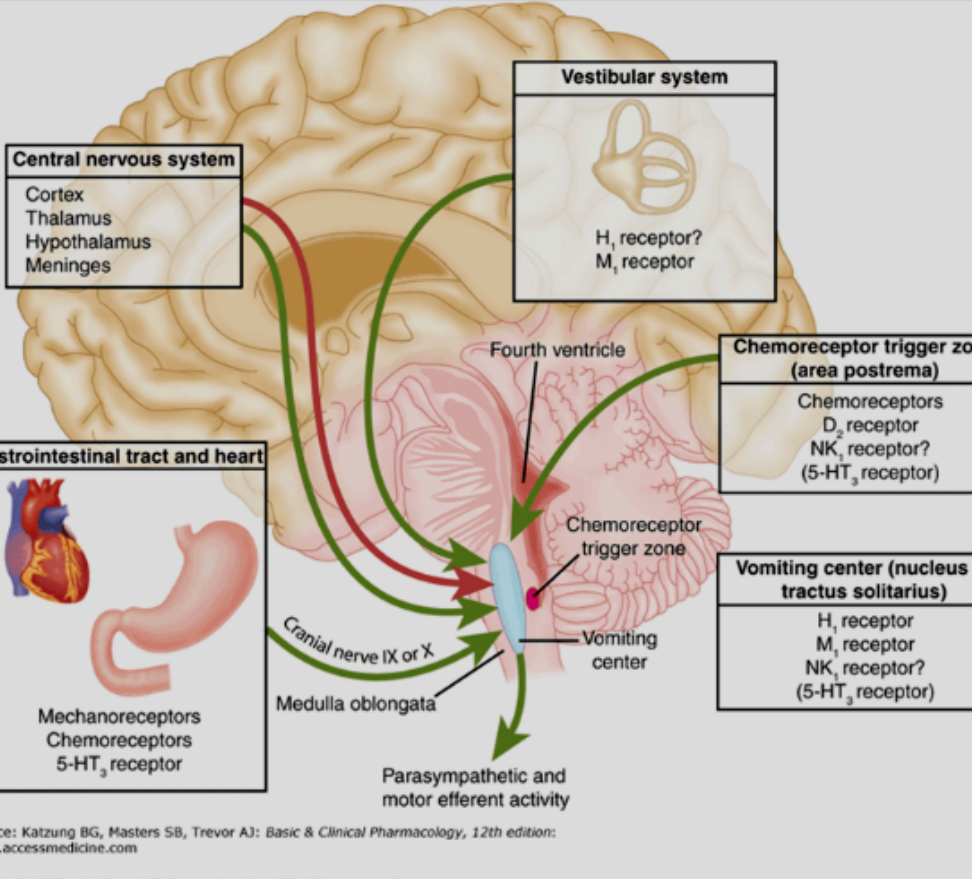
Is there another medication with a different mechanism you could add? Consider your answers as you read, and we'll revisit RW at the end of the brick.

GO TO CONCLUSION ↓

What Are Antiemetics?

The antiemetic medications are comprised of multiple different classes of medications that help reduce nausea and vomiting (i.e., anti - emesis). While nausea and vomiting can be a non-specific symptom of many different diseases and disorders it is commonly encountered in cancer patients. The cancer itself can be a stimulus within the chemo-receptor trigger zone (CTZ), but, more often, the chemotherapy drugs will induce receptors within the CTZ to cause nausea. This adverse reaction (i.e., nausea/ vomiting) is so predictable that we pretreat with medications.

The CTZ contains multiple receptors including D2 dopamine, 5-HT3 (serotonin), histamine-1, and neurokinin-1 (Figure 1). These all become important targets for medication therapy to treat chemotherapy induced nausea/ vomiting. Ma[illegible]cated in various areas throughout the body and medication therapy will target these receptors too in order to reduce nausea and vomiting. The classes of antiemetics that will be discussed in this brick include D2 dopamine receptor antagonists, 5-HT3 (serotonin) antagonists, neurokinin 1 (NK1) antagonists, antimuscarinics, and steroids.



Source: Katzung BG, Masters SB, Trevor AJ: *Basic & Clinical Pharmacology, 12th edition:* www.accessmedicine.com

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Figure 1

How do D2 Antagonists Work? (LO#1)

Metoclopramide is a D2 antagonist with many indications, but it also has many important adverse eff[illegible]ons come into play when metoclopramide[illegible]zone in the medulla (the area postrema). Blocking the D2 dopamine receptors in the area postrema inhibits stimulation of the vomiting center. It also acts as an antagonist at the 5-HT3 receptors, both centrally and vagally, increasing antiemetic action.

In the setting of chemotherapy induced nausea and vomiting the D2 antagonists are generally reserved for breakthrough nausea and vomiting, when other medications have not provided adequate relief, or in the setting of acute vomiting.

What are the adverse effects of D2 antagonists? (LO#2)

Metoclopramide must be used with caution, though, because the adverse effects can be severe. It can lead to restlessness, fatigue, and dystonia. Metoclopramide should not be used with selective serotonin reuptake inhibitors (SSRIs) because of the risk of serotonin syndrome, particularly in children and young adults and in high doses. There are also severe central nervous system (CNS) adverse effects, such as extrapyramidal symptoms, mental depression, tardive dyskinesia (often irreversible), Parkinsonian features, and neuroleptic malignant syndrome.

CLINICAL CORRELATION

Metoclopramide can increase milk production, i.e., cause lactation, in both males and females. This works beacuse dopamine naturally inhibits the secretion of prolactin, the hormone responsible for milk production. This 'adverse effect' can be targeted in a mother who is struggling to produce milk, however, given the other adverse effects of metoclopramide it is not used frequently for this indication.

What are the adverse effects of metoclopramide?

How do Serotonin (5–HT3) Antagonists Work? (LO#1)

Ondansetron, palonosetron, and granisetron are selective 5-HT3 antagonists. These antiemetics are able to block both the 5-HT3 receptors in the CTZ in the CNS and the 5-HT3 receptors on the vagus nerve terminals in the peripheral nervous system.

All of these medications are indicated for the treatment of nausea related to cancer chemotherapy. They are also recommended for prophylaxis against chemotherapy-induced nausea and vomiting. Chemotherapy can cause GI irritation, which in turn causes mucosal secretion of serotonin in the GI tract; this increases motility and leads to nausea.

What are the adverse effects of Serotonin (5–HT3) Antagonists? (LO#2)

Several adverse effects of these serotonin antagonists include headache, dizziness, and constipation. A major side effect to look out for is prolongation of the QT interval, but this is rare.

Mnemonic

Ondan**SEtRO**n, Palono**SEtRO**n, Grani**SEtRO**n

How do Neurokinin 1 Receptor Antagonists work? (LO#1)

Aprepitant is a neurokinin 1 (NK1) receptor antagonist (aka substance P antagonist) that is efficacious in the treatment of chemotherapy-induced nausea and vomiting. It acts by crossing the BBB and selectively blocking the NK1 receptor to inhibit the binding of substance P, which normally causes pain, nausea, and vomiting.

Because of the mecha[illegible]stance P by blocking NK1 receptors, this medication is not effective on an as needed basis. So, aprepitant is used to prevent chemotherapy induced nausea and vomiting. Patients will receive this medication as *pre-treatment,* i.e. ~30 min before, they receive their chemotherapy infusion.

What are the adverse effects of Neurokinin 1 Receptor Antagonists? (LO#2)

Aprepitant has few adverse effects compared to other antiemetic agents. Although uncommon, a couple of the adverse effects that are associated with this drug include neutropenia, which will be regularly monitored while the patient is receiving chemotherapy, and fatigue. However, the fatigue may also be closely associated with the chemotherapy the patient is receiving.

Mnemonic

To remember the action of aprepitant, think **AP**re**P**itant **A**ntagonizes substance **P**.

How do Antimusca[illegible]

The vestibular system communicates with the nucleus tractus solitarius in the medulla to induce nausea and vomiting. This vestibular nausea can be treated with antimuscarinics such as scopolamine. Scopolamine is a nonspecific muscarinic antagonist that crosses the BBB and blocks the M1 receptor to treat motion sickness, though they can be helpful with chemotherapy induced nausea and vomiting.

What are the adverse effects of antimuscarinics? (LO#2)

As we have previously discussed in detail, all anticholinergics have major adverse effects of pupillary dilation, dry mouth, and possible sedation (Recall the Mnemonic below!). These side effects are best avoided by using a transdermal patch.

Mnemonic

Remember adverse effects of muscarinic antagonists (blurry vision, overheating, dry mouth, flushed skin, urinary retention, and confusion) this way:
Blind as a Bat, **Hot** as a Hare, **Dry** as a Bone, **Red** as a Beet, **Full** as a Flask, **Mad** as a Hatter

How do Steroids work? (LO#1)

Corticosteroids, combined with other antiemetics, are commonly used as *pre-treatment* for chemotherapy induced nausea and vomiting. Their exact mechanism for their antiemetic properties is unknown, but it is thought to suppress the release of inflammatory mediators in the bloodstream that contribute to nausea. Dexamethasone, methylprednisone, and prednisone are the agents that are most commonly used.

What are the adverse effects of steroids? (LO#2)

Steroids can have many side effects, but when used for chemotherapy induced nausea and vomiting the most common are hyperglycemia and risk of GI bleed.

CLINICAL CORRELATION

Chemotherapy drugs can be categorized as high-risk for emesis (e.g., cisplatin, cyclophosphamide, doxorubicin), moderate-risk, or low-risk (e.g., most targeted MAbs). A multi-drug regimen is preferred for antiemetics to prevent stimuli from multiple sources. So, for example, a patient receiving cisplatin (highest emesis risk of the chemotherapy drugs with an incidence >90%) they would receive at least a three drug antiemetic regimen that includes: Aprepitant, ondansetron, and dexamethasone.

The three drug regimen would likely be just a starting point. They may have other medications like an antimuscarinic or D2 antagonist added, if needed.

Table 2 provides a summary of details about antiemetics.

Table 2

Type	Examples	Key Mechanism	Major Adverse Effects
D2 antagonists	Metoclopramide	Block D2 receptor to increase cholinergic stimulation of smooth muscle and increase gastric emptying for those with diabetic gastroparesis	QT prolongation, torsades de pointes, extrapyramidal symptoms, mental depression, tardive dyskinesia
5-HT3 antagonists	Ondansetron, granisetron	Block 5-HT3 receptors in area postrema	Constipation, diarrhea, headache, rare QT prolongation
NK1 antagonists	Aprepitant	Block NK1 receptor to inhibit substance P	Limited
Antimuscarinics	Scopolamine	Block M1 receptor to treat vertigo	Pupillary dilation, dry mouth, sedation
Steroids	Dexamethasone	Reduce inflammatory mediators	Hyperglycemia, GI bleed

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to RW, What treatment would you add?

RW currently has a prescription for ondansetron, a 5-HT3 receptor antagonist. There are, however, dopamine D2 antagonists, antimuscarinics, NK1 antagonists, and steroids. RW ne[illegible]o help with her nausea. Since the NK1 antagonists (aprepitant) and steroids (dexamethasone) are used primarily for pre-treatment before chemotherapy infusions we would likely not prescribe these in this instance. An antimuscarinic (scopolamine) could be a good option, however, they are most helpful for motion sickness. For this case, the best option would be to prescribe a dopamine D2 antagonist (metoclopramide) to help with RW's nausea.

Summary

- Antiemetics are medications used to reduce nausea and vomiting, commonly in cancer patients undergoing chemotherapy.
- The chemoreceptor trigger zone (CTZ) contains receptors like D2 dopamine, 5-HT3 serotonin, histamine-1, and neurokinin-1, which are targets for antiemetic therapy.
- Classes of antiemetics include D2 dopamine receptor antagonists, 5-HT3 serotonin antagonists, neurokinin 1 antagonists, antimuscarinics, and steroids.
- Metoclopramide, a D2 antagonist, has adverse effects such as restlessness, fatigue, dystonia, and serious CNS effects like tardive dyskinesia and neuroleptic malignant syndrome.
- 5-HT3 antagonists like ondansetron are used for chemotherapy-induced nausea, with side effects including headache and rare QT prolongation.
- Antimuscarinics like scopolamine treat motion sickness but can cause anticholinergic effects such as dry mouth and sedation.

Review Questic[illegible]

1. Which receptor does metoclopramide antagonize to increase antiemetic action?

A. ● D2 dopamine receptors
B. ○ 5-HT3 receptors
C. ○ Histamine-1 receptors
D. ○ NK-1 receptors

Hide Explanation

The correct answer is D2 dopamine receptors (A). Metoclopramide acts as a D2 antagonist, inhibiting stimulation of the vomiting center. It also acts on 5-HT3 receptors (B), but the primary antiemetic action is through D2 antagonism. Histamine-1 receptors (C) and NK1 receptors (D) are not the primary targets of metoclopramide.

2. Which is a major side effect of serotonin antagonists?

A. ○ Headache
B. ○ Dizziness
C. ● QT interv[illegible]
D. ○ Constipation
E. ○ GI irritation

Hide Explanation

The correct answer is QT interval prolongation (C). This is a rare but significant side effect of serotonin antagonists. Headache (A), dizziness (B), and constipation (D) are common side effects but not considered major. GI irritation (E) is related to chemotherapy, not the medication.

3. Which receptor does aprepitant block to treat nausea?

A. ○ Serotonin receptor
B. ● NK1 receptor
C. ○ Dopamine receptor
D. ○ Histamine receptor
E. ○ GABA receptor

Hide Explanation

The correct answer is[illegible]in 1 (NK1) receptor

The correct answer is NK1 receptor (B). Aprepitant is a neurokinin 1 (NK1) receptor antagonist, blocking substance P binding to treat nausea. Serotonin receptor (A) antagonists are used for nausea but not aprepitant. Dopamine receptor (C) antagonists treat nausea but differ from aprepitant. Histamine receptor (D) antagonists are used for allergies, not nausea. GABA receptor (E) is unrelated to nausea treatment.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of the antiemetic agents.~~

02 ☑ ~~Identify adverse effects associated with the antiemetic agents.~~

Objective Confidence: 2/2